

Product Microservice - REST API Reference

Complete API Endpoint Specification, Request Payloads & Response References

Base URL: `http://localhost:8083/api/v1/products` | Port: 8083 | Auth: JWT Bearer Header

API Endpoints Summary Matrix

METHOD	ENDPOINT URL	AUTHENTICATION REQUIRED	DESCRIPTION
POST	/api/v1/products	Yes (ADMIN / SELLER)	Create a new product
GET	/api/v1/products	No (Public)	Get all products list
GET	/api/v1/products/{id}	No (Public)	Get product details by ID
PUT	/api/v1/products/{id}	Yes (ADMIN / SELLER)	Update existing product
DELETE	/api/v1/products/{id}	Yes (ADMIN)	Delete product by ID
GET	/api/v1/products/search	No (Public)	Search products by keyword
GET	/api/v1/products/category/{category}	No (Public)	Get products by category

Detailed API Endpoint Specifications

POST

/api/v1/products

1. Create Product

Description: Adds a new product record to the database.

Headers: Authorization: Bearer <JWT_TOKEN> | Content-Type: application/json

REQUEST BODY (JSON):

```
{
  "name": "Wireless Noise Cancelling Headphones",
  "brand": "AudioPro",
  "category": "Electronics",
  "description": "High fidelity over-ear bluetooth headphones with active noise cancellation.",
  "price": 199.99,
  "stock": 50,
  "seller": "TechWorld Store",
  "imageUrl": "https://images.example.com/products/headphones.jpg",
  "active": true
}
```

SUCCESS RESPONSE (HTTP 201 CREATED):

```
{
  "id": 1,
  "name": "Wireless Noise Cancelling Headphones",
  "brand": "AudioPro",
  "category": "Electronics",
  "description": "High fidelity over-ear bluetooth headphones with active noise cancellation.",
  "price": 199.99,
  "stock": 50,
  "seller": "TechWorld Store",
  "imageUrl": "https://images.example.com/products/headphones.jpg",
  "active": true
}
```

ERROR RESPONSE (HTTP 400 BAD REQUEST):

```
{
  "timestamp": "2026-08-17T00:30:00",
  "status": 400,
  "error": "Bad Request",
  "message": "Validation failed for request object",
  "path": "/api/v1/products",
  "validationErrors": {
    "name": "Product name is required",
    "price": "Price must be greater than 0"
  }
}
```

GET

/api/v1/products

2. Get All Products

Description: Returns list of all available products.

Headers: None (Public Endpoint)

REQUEST BODY:

None

SUCCESS RESPONSE (HTTP 200 OK):

```
[
  {
    "id": 1,
    "name": "Wireless Noise Cancelling Headphones",
    "brand": "AudioPro",
    "category": "Electronics",
    "description": "High fidelity over-ear bluetooth headphones.",
    "price": 199.99,
    "stock": 50,
    "seller": "TechWorld Store",
    "imageUrl": "https://images.example.com/products/headphones.jpg",
    "active": true
  },
  {
    "id": 2,
    "name": "Ergonomic Gaming Mouse",
    "brand": "LogiTech",
    "category": "Electronics",
    "description": "Precision optical sensor mouse.",
    "price": 79.50,
    "stock": 120,
    "seller": "GamingGear",
    "imageUrl": "https://images.example.com/products/mouse.jpg",
    "active": true
  }
]
```

GET

/api/v1/products/{id}

3. Get Product By ID

Description: Fetches single product details by ID.

Path Variable: id (Long, e.g. 1)

SUCCESS RESPONSE (HTTP 200 OK):

```
{
  "id": 1,
  "name": "Wireless Noise Cancelling Headphones",
  "brand": "AudioPro",
  "category": "Electronics",
  "description": "High fidelity over-ear bluetooth headphones.",
  "price": 199.99,
  "stock": 50,
  "seller": "TechWorld Store",
  "imageUrl": "https://images.example.com/products/headphones.jpg",
  "active": true
}
```

ERROR RESPONSE (HTTP 404 NOT FOUND):

```
{
  "timestamp": "2026-08-17T00:30:00",
  "status": 404,
  "error": "Not Found",
  "message": "Product not found with ID: 999",
  "path": "/api/v1/products/999"
}
```

PUT**/api/v1/products/{id}**

4. Update Product

Description: Updates fields of an existing product.

Path Variable: id (Long, e.g. 1) | **Headers:** Authorization: Bearer <JWT_TOKEN>

REQUEST BODY (JSON):

```
{
  "name": "Wireless Noise Cancelling Headphones v2",
  "brand": "AudioPro",
  "category": "Electronics",
  "description": "Updated version with 40h battery life.",
  "price": 179.99,
  "stock": 45,
  "seller": "TechWorld Store",
  "imageUrl": "https://images.example.com/products/headphones-v2.jpg",
  "active": true
}
```

SUCCESS RESPONSE (HTTP 200 OK):

```
{
  "id": 1,
  "name": "Wireless Noise Cancelling Headphones v2",
  "brand": "AudioPro",
  "category": "Electronics",
  "description": "Updated version with 40h battery life.",
  "price": 179.99,
  "stock": 45,
  "seller": "TechWorld Store",
  "imageUrl": "https://images.example.com/products/headphones-v2.jpg",
  "active": true
}
```

DELETE**/api/v1/products/{id}**

5. Delete Product

Description: Deletes product by ID.

Path Variable: id (Long, e.g. 1) | **Headers:** Authorization: Bearer <JWT_TOKEN>

SUCCESS RESPONSE (HTTP 204 NO CONTENT):

Empty Body (Status Code 204)

ERROR RESPONSE (HTTP 404 NOT FOUND):

```
{
  "timestamp": "2026-08-17T00:30:00",
  "status": 404,
  "error": "Not Found",
  "message": "Product not found with ID: 999",
  "path": "/api/v1/products/999"
}
```

GET

</api/v1/products/search?keyword={keyword}>

6. Search Products

Description: Searches product catalog by keyword matching product name.

Query Parameter: keyword (String, e.g. Mouse)

SUCCESS RESPONSE (HTTP 200 OK):

```
[
  {
    "id": 2,
    "name": "Ergonomic Gaming Mouse",
    "brand": "LogiTech",
    "category": "Electronics",
    "description": "Precision optical sensor mouse.",
    "price": 79.50,
    "stock": 120,
    "seller": "GamingGear",
    "imageUrl": "https://images.example.com/products/mouse.jpg",
    "active": true
  }
]
```

GET

</api/v1/products/category/{category}>

7. Get Products By Category

Description: Filters product catalog matching specified category.

Path Variable: category (String, e.g. Electronics)

SUCCESS RESPONSE (HTTP 200 OK):

```
[
  {
    "id": 1,
    "name": "Wireless Noise Cancelling Headphones",
    "brand": "AudioPro",
    "category": "Electronics",
    "price": 199.99,
    "stock": 50,
    "active": true
  },
  {
    "id": 2,
    "name": "Ergonomic Gaming Mouse",
    "brand": "LogiTech",
    "category": "Electronics",
    "price": 79.50,
    "stock": 120,
    "active": true
  }
]
```